
Require: Token stream σ from the LLM; context-free grammar G of the target language.

Ensure: A sequence of execution outputs, yielded incrementally;
PARSEERROR or EXECUTIONERROR raised on the first failure.

Invariant. At the start of every iteration of the main loop, b contains exactly the suffix of the tokens emitted so far that has not yet been dispatched for execution.

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1: procedure JitGen( $\sigma, G$ )
2:    $\iota \leftarrow \text{CREATEINTERPRETER}()$ 
3:    $b \leftarrow \varepsilon$  ▷ unexecuted-suffix buffer
4:   for all  $f \in \sigma$  do
5:      $b \leftarrow b \cdot f$ 
6:      $(\tau, ok) \leftarrow \text{PARSE}(G, b)$ 
▷ returns  $ok=\text{INCOMPLETE}$  or raises  $\text{PARSEERROR}$ 
7:     if  $ok = \text{INCOMPLETE}$  then
8:       continue
9:     end if
10:     $S \leftarrow \text{TOPLEVELSTATEMENTS}(\tau)$ 
11:    if  $|S| \geq 2$  then
12:      Stability:  $S[1 : |S|-1]$  cannot be extended by future tokens.
13:      for  $i \leftarrow 1$  to  $|S| - 1$  do
14:         $c \leftarrow b[ S[i].start .. S[i].end ]$ 
15:         $o \leftarrow \iota.\text{EXECUTE}(c)$ 
16:        yield  $o$ 
17:      end for
18:       $b \leftarrow b[ S[|S|].start .. ]$ 
19:    end if
20:  end for
21:  end procedure
```
